| | Hypothesis of the growth theory | Outcome | Measured |
|---|---|---|---|
| **H3** | Recovery loop collapses into one multiplier | Confirmed | slope 1.24, $R^2$ 0.999 |
| **H4** | In-use and discard respond with opposite sign | Confirmed | Spearman −0.95 |
| **H5** | General-equilibrium amplification negligible | Confirmed | amplification 1.0145 |
| **H11** | Short-run closed form for $\partial \ln s / \partial \rho$ | Confirmed | 0.76 vs 0.74 |
| **H2** | Faster growth dilutes recovery | Sign right, size wrong | net −0.38 |
| **H7** | Intensity of use constant | Refined | elasticity 0.71 |
| **H1** | Coefficients invariant to the economy | Contradicted | 0.33 of cells respond |
| **H6** | Energy rations throughput proportionally | Contradicted | $\partial \chi$/energy +0.10 per SD |
| **H8** | Balanced growth path | Contradicted | elasticities 0.87/0.71/1.07 |
| **H9** | Substitutability $\pi$ unidentifiable from tonnage | Extension | $\pi$ range 0.019 < 95% interval 0.072 |
| **H10** | Recovery on saturated Hill branch | Extension | all years; min $m/m_{\text{infl}}$ 1.49 |